WORKED SOLUTION

# STEP II 2003, Question 4

Pure • Circle Segment Areas

**STEP II 2003, Question 4**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q4

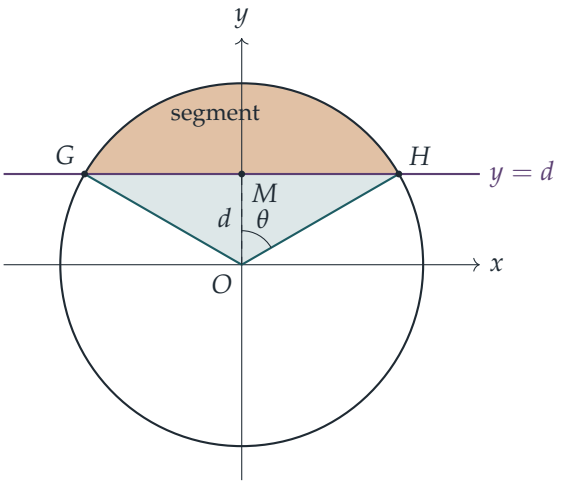
## The key idea

A minor segment is a sector with a triangle removed, so the first result is just "sector minus triangle". For the rest, notice that in $(*)$ the number $d$ is really the *perpendicular distance from the centre of the circle to the line*. Sliding or rotating the picture changes no areas, so in every case we keep $(*)$ and replace $d$ by that distance. All the work is in computing the distance.

## Solution

### The basic result

Because the line meets the circle at two points, $0 < d < R$. Putting $y = d$ into $x^2 + y^2 = R^2$ gives $x = \pm\sqrt{R^2 - d^2}$, so $G = (-\sqrt{R^2 - d^2}, d)$ and $H = (\sqrt{R^2 - d^2}, d)$. Let $M = (0, d)$ be the midpoint of $GH$, and let $\theta = \angle MOH$.



In the right-angled triangle $OMH$ the hypotenuse is $OH = R$ and the adjacent side is $OM = d$, so $\cos\theta = d/R$. Since $0 < d/R < 1$, we have $\theta = \arccos(d/R)$ with $0 < \theta < \frac{\pi}{2}$, and by symmetry $\angle GOH = 2\theta < \pi$.

The sector $OGH$ with angle $2\theta$ has area

$$\tfrac{1}{2}R^2(2\theta) = R^2 \arccos\left(\frac{d}{R}\right).$$

Triangle $OGH$ has base $GH = 2\sqrt{R^2 - d^2}$ and height $OM = d$, so its area is $d\sqrt{R^2 - d^2}$. Because $2\theta < \pi$, the centre $O$ lies on the opposite side of $GH$ from the minor arc, so the sector is exactly the triangle together with the minor segment. Hence

$$\text{area of minor segment} = R^2 \arccos\left(\frac{d}{R}\right) - d\sqrt{R^2 - d^2},$$

which is $(*)$.

**Ask yourself** What does $d$ actually stand for in $(*)$, as far as the geometry is concerned?

Nothing in the argument used the axes: it used only the radius $R$ and the distance $OM = d$ from the centre to the chord. Translating or rotating the whole figure preserves both, and preserves area. So for any circle of radius $R$ and any line at perpendicular distance $p$ from the centre, with $0 \le p < R$, the minor segment has area

$$R^2 \arccos\left(\frac{p}{R}\right) - p\sqrt{R^2 - p^2}. \qquad (\dagger)$$

(If $p = 0$ the line is a diameter, the two pieces are equal halves, and $(\dagger)$ gives $\frac{1}{2}\pi R^2$ correctly.) In each part we find $p$ and substitute it into $(\dagger)$.

#### Part (i)

The centre is $(a, b)$ and the line $y = c$ is horizontal, so the distance from the centre to the line is $p = |c - b|$. Replace $d$ by $|c - b|$:

$$R^2 \arccos\left(\frac{|c-b|}{R}\right) - |c-b|\sqrt{R^2 - (c-b)^2}.$$

The modulus matters: $c$ may be below $b$.

#### Part (ii)

Here the centre is $O$. We need the distance from $O$ to $y = mx + c$. Let the line make angle $\alpha$ with the $x$-axis, so $\tan\alpha = m$, and let it cross the $y$-axis at $C = (0, c)$. Drop the perpendicular from $O$ to the line, meeting it at $F$. The angle between $OC$ (vertical) and $OF$ (perpendicular to the line) equals the angle between the horizontal and the line, namely $\alpha$. So in the right-angled triangle $OFC$,

$$p = OF = |c|\,|\cos\alpha| = \frac{|c|}{\sqrt{1 + \tan^2\alpha}} = \frac{|c|}{\sqrt{1 + m^2}}.$$

(This is the usual formula $|Ax_0 + By_0 + C|/\sqrt{A^2+B^2}$ for the line $mx - y + c = 0$ and the point $(0,0)$.) Then

$$\sqrt{R^2 - p^2} = \sqrt{R^2 - \frac{c^2}{1+m^2}} = \frac{\sqrt{R^2(1+m^2) - c^2}}{\sqrt{1+m^2}},$$

so the area of the minor segment is

$$R^2 \arccos\left(\frac{|c|}{R\sqrt{1+m^2}}\right) - \frac{|c|\sqrt{R^2(1+m^2)-c^2}}{1+m^2}.$$

The line meets the circle exactly when $c^2 < R^2(1+m^2)$, so the square root is real.

#### Part (iii)

Shift the origin to the centre by writing $X = x - a$, $Y = y - b$. The circle becomes $X^2 + Y^2 = R^2$ and the line becomes

$$Y + b = m(X + a) + c \quad \Longleftrightarrow \quad Y = mX + k, \qquad k = c + ma - b.$$

This is part (ii) with $c$ replaced by $k$, and translation does not change area. So $p = |c + ma - b|/\sqrt{1+m^2}$, and the area of the minor segment is

$$R^2 \arccos\left(\frac{|c+ma-b|}{R\sqrt{1+m^2}}\right) - \frac{|c+ma-b|\sqrt{R^2(1+m^2)-(c+ma-b)^2}}{1+m^2}.$$

Putting $m = 0$ recovers part (i), and putting $a = b = 0$ recovers part (ii).

**?** **Ask yourself** Why can't we just write $c - b$ in part (i), without the modulus?

If $c < b$ then $(c-b)/R$ is negative, arccos returns an angle $\pi - \theta$ greater than $\frac{\pi}{2}$, and the second term changes sign. The expression becomes $R^2(\pi - \theta) + p\sqrt{R^2 - p^2}$, which is the area of the *major* segment, not the minor one.

## Checks

- With $R = 2$, $d = 1$ we get $\theta = \frac{\pi}{3}$, and $(*)$ gives $\frac{4\pi}{3} - \sqrt{3} \approx 2.4567$. This is less than half the disc ($2\pi \approx 6.28$), as it should be.
- As $d \to R$ the area tends to 0, and at $d = 0$ it equals $\frac{1}{2}\pi R^2$.
- We checked all four formulas against direct numerical integration of chord lengths over hundreds of random circles and lines. The agreement was within $2 \times 10^{-6}$.

## Common slips

- Leaving out the modulus. A signed distance gives the major segment whenever the centre is on the "wrong" side of the line.

- Using the vertical gap $|c|$ instead of the perpendicular distance $|c|/\sqrt{1+m^2}$ in parts (ii) and (iii).
- Using $\frac{1}{2}R^2\theta$ for the sector, forgetting that the angle at the centre is $2\theta$, not $\theta$.

## Beyond STEP

The question rests on the fact that area is unchanged by *isometries*, the translations, rotations and reflections of the plane. In a first-year course this becomes the statement that the Jacobian determinant of an isometry has modulus 1, so the change-of-variables formula for double integrals leaves areas unchanged. Classifying isometries, and the quantities they preserve, is a good route into geometry and group theory. Strang's *Introduction to Linear Algebra* covers rotation matrices and determinants as scale factors for area.

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